

Part Two: Take-Home Examination

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Moderator: Mr RQ Odendaal

Release Date: 25 July 2020 at 11:00
Due Date: 25 July 2020 at 12:00

Please read and understand the following.

1. All test and examination regulations of the University of Pretoria apply.
2. This is an OPEN BOOK assignment. You may use any source available to you, like your textbook and lecture notes, to assist in answering the questions. However, you **MAY NOT** receive (or provide) help in any way whatsoever from (or to) any other person. This includes the lecturer, tutor, a friend and family members.
3. Answer the questions **in your own handwriting** on separate paper, lined or unlined.
Take quality photographs of your answer sheets and scan them as a PDF document. Use apps like **CamScanner** to convert the photographs to PDF.
Submit your answer sheets, via ClickUP, in the “**FSK116 Sick/Aegrotat and Supplementary Exam**” folder under the “**ONLINE**” headline. **We urge you to submit before 12:00.**
There will be a queue during submission. You may not get your submission receipt immediately! To accommodate this, we give you a grace period of 30 minutes only between 12:00 and 12:30 to make your submission.
YOU HAVE BEEN WARNED: Late submissions will not be accepted!
4. Number and label your submissions as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
5. Provide neatly labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct numbers of significant figures, units, etc. Failure to do so will result in marks being deducted.
6. Your answer needs to be written in a systematic and logical fashion. Map out the steps you will take to reach your goal i.e. solve the problem in your mind first! You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not necessarily mean that you will receive full marks!
7. Units must be included when numbers are substituted; else, marks will be deducted.
8. You will need to look up constants and typical values for parameters in your calculations.
9. If any misconduct with respect to this examination is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.

DECLARATION OF ORIGINALITY

UNIVERSITY OF PRETORIA - Department of Physics

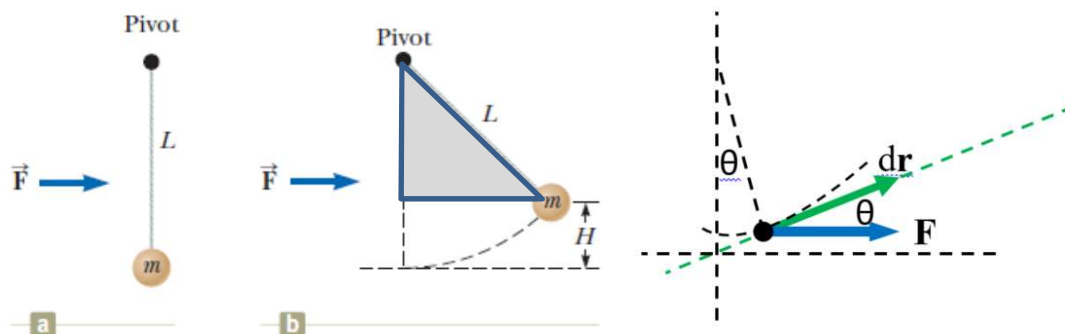
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I understand what plagiarism is and am aware of the University's policy in this regard. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This statement is a reminder of the Code of Conduct that you have already signed.

TOTAL POINTS: 27 for Supplementary exam

QUESTION 1 [16]

As shown in the figure below, a ball is suspended vertically (at rest) from a pivot point by a strong lightweight chord. When a wind of constant force $\vec{F}$ is incident on the ball, it swings upward to some height above the starting point, as seen in the figure below. The height H can be obtained as a function of the magnitude of $\vec{F}$. We will work through this process; and, we start by assuming that the ball never attains an angle of 90° with the vertical. The sketch on the far right of the figure below indicates an instant during the ball's motion along the curved path (it is drawn out of scale). Take a minute to read all of the sub-questions to get a full picture, there are clues scattered all over.



(a) Assuming the system we will use for analysis consists of the ball and the Earth. Is this system isolated or non-isolated? (1)

(b) Considering the system, draw a free-body diagram indicating the two forces that are acting on the ball and are external to it? (2)

(c) State the energy conservation equation that is appropriate for this system. It has to be specific to this system and not just a general conservation principle. (3)

(d) (i) Write the expressions for the work done by the external forces as dot products. (2)

HINT: These expressions should be in integral form. You can use generic " $\mathbf{r}_{\text{initial}}$ " and " $\mathbf{r}_{\text{final}}$ " to indicate the integration bounds. (2)

(ii) Which of these work done integrals become zero. Explain why. (2)

(e) From the equations in (c) and (d), solve for H . You need an appropriate coordinate axes set. (2)

(f) From the grey shaded triangle in the middle figure, one could use the Pythagorean Theorem to write that:

$$x^2 + (L - H)^2 = L^2$$

Solve for x , the horizontal displacement of the ball, in this expression. Indicate the appropriate location for a coordinate axes (frame of reference) for this problem. (2)

(g) Show that H can be written as: (2)

$$H = \frac{2L}{1 + \left(\frac{mg}{F}\right)^2}$$

QUESTION 2 [11]

Steam at $100\text{ }^{\circ}\text{C}$ is added to ice at $0\text{ }^{\circ}\text{C}$.

- (a) Find the amount of ice melted and the final temperature when the mass of steam is 14.0 g and the mass of ice is 47.0 g . (7)
- (b) Repeat this calculation, when the mass of steam is 1.40 g and the mass of ice is 47.0 g . (4)

You will need to look up some parameter values in the textbook or elsewhere. You do not need to reference the sources.

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